

The impact of polyurethane adhesives on the recyclability of polyethylene-based laminated films

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ABSTRACT

Laminated plastic films, especially mono-material multilayer polyethylene-based (MM-PE) films, have attracted strong interest within the packaging industry over the past decade. Towards closing the material loop, MM-PE films should be recycled in a film-to-film mechanical recycling process, while preserving their desired performance. However, MM-PE laminates typically contain polyurethane (PUR) adhesives between PE film layers, which can affect their recyclability. Herein, three MM-PE films containing different PUR adhesives were studied to assess the effect of the PUR characteristics on the films' recyclability. The processability of MM-PE laminated films produced via film-blowing extrusion was evaluated, followed by structural, rheological, mechanical, and optical characterization of the recycled films. The composition of the PUR adhesive substantially affects the closed-loop recyclability of the films, and even a small PUR content (2 wt%) can hinder the recycling process. Overall, PUR adhesives reduce the zero-shear viscosity and increase the heterogeneity of the recycled laminates, reducing toughness and increasing haziness. By assessing the morphology and the physicochemical characteristics of the PUR adhesives, it is possible to estimate the impact of these materials on the recyclability of MMPE laminates. PURs presenting a higher degree of mixing between hard and soft phases, as indicated by a lower content of hydrogen-bonded groups among hard segments, are preferred for improving the recyclability of MM-PE laminates.

1. Introduction

Each year, 21 million tonnes of plastic packaging are produced in the European Union (EU) [1], of which approximately 23 % consists of polyethylene (PE), a family of polymers that includes low-density polyethylene (LDPE), linear low-density polyethylene (LLDPE), and high-density polyethylene (HDPE) [2]. According to the EU's Strategy for Plastic in a Circular Economy, by 2030, all plastic packaging placed on the EU market should be cost-effectively reusable or recyclable [3]. Therefore, the design for recycling of PE-based packaging products has been receiving significant academic and industrial attention [4–6].

An important segment of packaging is PE flexible packaging, commonly referred to as plastic film. These plastic films can be categorized into monolayer (one layer of one plastic type), mono-material multilayer (several layers, but only one plastic type), or multi-material multilayer (several layers, several plastic types) [6]. To meet the packaging industry requirements, including mechanical properties, gas

barrier, optical properties, sealing, and chemical resistance, multi-material multilayer films combine PE with different polymers, including polypropylene (PP), polyethylene terephthalate (PET), polyamide (PA), and poly(ethylene-vinyl alcohol) (EVOH) [7–9]. Due to the complexity of these multi-material multilayer films, their mechanical recycling is challenging [10].

The environmental performance of mono- and multi-materials plastic films from the packaging industry, when assessed by life cycle assessment (LCA), is heavily influenced by their composition [11]. Only the mono-materials packaging systems are currently considered as fully recyclable [11–13].

In certain applications, including dry food and detergent packaging, mono-material multilayer flexible films can replace the multi-material ones, which may lead to improved recyclability of the films [6,14]. PE-based mono-material multilayer films (MM-PE) are suitable candidates for mechanical recycling, as PE is used across all layers [6,12]. Recently, special grades of LLDPE and HDPE have been developed for

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machine-direction orientation (MDO) after blown or cast extrusion [14, 15]. Due to the development of these new oriented PE films (MDOPE) with different molecular architectures and morphologies, PE-based layers can offer higher stiffness, puncture resistance, and better optical properties, making them suitable for the replacement of PET in multi-material multilayers and developing MM-PE laminates instead [15,16].

Depending on the packaging requirements, multilayer flexible films are produced through co-extrusion or lamination [6]. Since the production processes for MDOPE and LDPE layers differ, the MM-PE films are produced using a lamination process, and PUR adhesives are commonly used as a binder in the final laminate structure of the multilayer flexible film [17]. PUR is synthesized from the reaction of isocyanates and polyols. Isocyanate (-NCO) groups undergo addition reactions with hydrogen-containing nucleophiles, such as hydroxyl (-OH), amino (-NH_2), or carboxyl (-COOH) groups [18]. PUR adhesives are generally formed from the reaction of an NCO-terminated prepolymer with a polyester- or polyetherpolyol [19–23]. Common monomers for polyester synthesis include terephthalic acid (TPA), isophthalic acid (m-phthalic acid), adipic acid, diethylene glycol, neopentyl glycol, and ethylene glycol [24]. Among polyethers, polypropylene glycol (PPG) is one of the most widely used polyols for adhesive formulations [25].

Both aromatic and aliphatic diisocyanates are employed in the PUR adhesive synthesis. Typical aromatic isocyanates include toluene diisocyanate (TDI) and 4,4'-diphenylmethane diisocyanate (MDI) [26], while widely used aliphatic options are hexamethylene diisocyanate (HDI), isophorone diisocyanate (IPDI), and xylylene diisocyanate (XDI) [27, 28]. The isocyanate index, defined as the molar ratio of NCO to OH groups, is commonly in the range of 1.5–2 to achieve an optimal balance between reactivity, properties, and processing requirements, and results in a crosslinked PUR adhesive [19,21].

The choice of polyol backbone, the type of the diisocyanate, and the isocyanate index represent the key parameters affecting the performance of PUR laminating adhesives, including flexibility, adhesion strength, chemical resistance, heat stability, and retortability [18,19, 28]. The final structure and morphology of PUR adhesives depend on the chemical structure and molecular weight of the raw materials, the volume fraction of the hard and soft segments, and the extent of intra- and inter-molecular interactions in each segment and between hard and soft segments, especially hydrogen bond formation [29–34]. Additionally, the content of PUR adhesive used during film lamination affects the performance of the adhesive [35,36].

In line with the principles of the circular economy, and to reduce dependence on virgin plastics while minimizing environmental impacts [37,38], the recycled MM-PE ideally will be reused in the same application as the original product, or in applications that demand comparable material properties [37,39]. Currently, mechanical recycling remains the dominant method for recycling of polyolefins [40]. Achieving closed-loop film-to-film mechanical recycling of MM-PE requires that the recycled material be processable through film-blowing, one of the most widely used polymer processing technologies for the production of plastic films [41].

In general, the processability and performance of a final film-blown product are highly dependent on the macromolecular architecture of PE, particularly its molecular weight, molecular weight distribution, and the content and distribution of short-chain branches [42]. However, MM-PE films typically contain PUR adhesives, and due to the presence of polar chemical groups in PUR, such as urethane and ester/ether functionalities, there is an inherent incompatibility of PUR with the non-polar PE matrix during the melt blending stage of mechanical recycling [43,44]. It has been previously found that contaminations will drastically reduce the processing window and the quality of the final film [45]. Consequently, it is essential to investigate the influence of the chemical structure of PUR adhesive on the processability and final performance of the recycled films within the closed-loop mechanical recycling of MM-PE.

The present study systematically evaluates the recyclability of MM-

PE laminated films. Herein, the impact of three different types of PUR adhesives on the processability, mechanical, and optical performance of MM-PE laminates is evaluated in a closed-loop film-to-film mechanical recycling process. To the best of our knowledge, no comprehensive study currently exists for evaluating the recyclability of MM-PE laminates containing PUR adhesives. This study aims to identify the main effects of PUR adhesives on the film-blowing processability and properties of recycled MM-PE films.

2. Materials and methods

2.1. Mono-material multilayer polyethylene-based (MM-PE) laminates

Three commercial MM-PE laminates, each one containing a different PUR adhesive (PU1, PU2, and PU3) were evaluated. The composition of the layers in all the laminates was consistent, comprising a 50 μm MDOPE layer, a 2 g/m^2 PUR adhesive, and a 50 μm LDPE film as a sealant layer. A film consisting of MDOPE/LDPE without PUR adhesive (PU0) was used as a control.

2.2. Closed-loop mechanical recycling

In this study, closed-loop mechanical film to film recycling refers to the process of converting MMPE laminated films into new films through a shredding, compounding and film-blowing process. At the first step, the laminates were comminuted in a shredder with a 6 mm diameter sieve (Hellweg MDS 340/150). Before the compounding and granulation steps, the shredded flakes were dried for 24 h at 60 °C. The dried flakes were fed into a co-rotating twin screw extruder compounder (Krauss-Maffei ZE 25R52D) at temperatures ranging from 30 to 230 °C (feeding zone to the die), using both vacuum and atmospheric degassing. The screw rotation speed was maintained at 270 rpm. The resulting pellets were dried again (1 h at 60 °C) before proceeding to the film-blowing step.

The film-blowing process was conducted using a Brabender single-screw extruder (screw diameter 30 mm; length-to-diameter, $L/D = 25$) equipped with a melt pressure sensor and a screen changer (100 mesh). The channel depth of the screw in the metering section was 1.66 mm. The film-blowing die had a diameter of 40 mm and a die gap of 1 mm. The barrel temperature was set in the range of 190–220 °C, and the die head was maintained at 220 °C. Two different screw speeds, 25 and 40 rpm, were set during the production of the film. The cooling fan rate was adjusted for all samples to keep the frost line height (FLH) at 12.5 cm. The blow-up ratio (BUR) and draw-down ratio (DDR) were calculated using Equations (1) and (2), respectively:

$$BUR = \frac{D_B}{D_D} \quad (1)$$

$$BDR = \frac{h}{t \times BUR} \quad (2)$$

Where D_B is the bubble diameter, D_D the die diameter, h the die gap, and t is the film thickness.

The processing for each sample was conducted at two different film thicknesses (25 and 55 μm), corresponding to two different DDRs. A 25 μm thickness setup was selected as the critical benchmark for evaluating processability, as thinner films present more challenges in maintaining a stable bubble.

During processing. Afterward, 55 μm thickness films were produced since thicker films typically present a lower coefficient of variation in the film thickness. These films were then employed for measuring the mechanical and optical properties, as well as for the analysis of the gel particles through optical microscopy. Table A.2 in the supplementary information (SI) summarizes the film-blowing processing parameters for both film thicknesses setups. Each experimental run was replicated twice to estimate the variances. The processability of recycled MM-PE

laminates was evaluated by assessing the average machine load, average melt pressure, and melt pressure build-up (MPBU) (Equation (3)) in the filmblowing extrusion section during the 25 μm film setup based on 2 runs of 30 min each per sample.

$$MPBU = \frac{P_f - P_i}{P_{ave}} \times 100 \quad (3)$$

Where P_i and P_f are the melt pressure at the start and end of processing, respectively. P_{ave} is the average melt pressure during the process. In addition to these parameters, the number of width reductions (indicating a situation where there is bubble instability, resulting in a reduction in the film width – Fig. 1b), and the number of bubble tearing (where the bubble tears completely, requiring the process to be restarted – Fig. 1c) were evaluated.

2.3. Characterization of the PE film layers, PUR, and recycled MM-PE after compounding

Differential scanning calorimetry (DSC), melt flow rate (MFR), density, and rheological analysis of the laminate layers were performed on compression-molded sheets, which were prepared directly from discarded shredded films. The compression molding was carried out at 220 $^{\circ}\text{C}$ for 5 min.

2.3.1. Differential scanning calorimetry (DSC)

Prior to the DSC analysis of PUR adhesives, a reflux technique was employed to dissolve the PE layers of the laminates. For this extraction, approximately 10 g of MM-PE films were placed in a 500 mL round-bottom flask containing 200 mL of p-xylene at 150 $^{\circ}\text{C}$ and kept under reflux for 15 min. After cooling to room-temperature, the sample was filtered to separate the PUR adhesive from the PE solution. The recovered PUR was further washed with fresh p-xylene to remove residual PE, and subsequently dried under 10 mbar vacuum at 40 $^{\circ}\text{C}$ for 48h. DSC measurements of the extracted PUR adhesives were conducted on a DSC Q2000 (TA Instruments). All DSC measurements were performed under nitrogen atmosphere, using samples of ca. 8 mg. The heating and cooling rates were set to 30 $^{\circ}\text{C}/\text{min}$. The PUR samples were heated from –60 to 120 $^{\circ}\text{C}$, kept at 120 $^{\circ}\text{C}$ for 3 min, cooled down to –60 $^{\circ}\text{C}$, and then re-heated to 120 $^{\circ}\text{C}$.

DSC measurements of the lamination PE layers (MDOPE and LDPE) and the control film (PU0) were conducted on a DSC 300 Caliris (Netzsch). All DSC measurements were performed under nitrogen atmosphere, using samples of ca. 8 mg. The heating and cooling rates were set to 10 $^{\circ}\text{C}/\text{min}$. The samples were heated from 0 to 160 $^{\circ}\text{C}$, following an isothermal step for 10 min, cooled to 0 $^{\circ}\text{C}$, maintained at this

temperature for 10 min, and then re-heated to 160 $^{\circ}\text{C}$. The crystallinity degree (X_c) was calculated according to Equation (4), where ΔH_m is the measured melt enthalpy, ΔH_m^{∞} is the theoretical enthalpy of the 100 % crystalline PE (293 J/g) [46].

$$X_c(\%) = \frac{\Delta H_m}{\Delta H_m^{\infty}} \times 100 \quad (4)$$

2.3.2. Melt flow rate (MFR)

The melt flow rate (MFR) of the pure layers (MDOPE and LDPE) was measured in accordance with ISO 1133–1:2022. Measurements were carried out using a melt flow indexer (Tinius Olsen MP1200) at a temperature of 190 $^{\circ}\text{C}$ under a load of 2.16 kg.

2.3.3. Density

The density of PE layers and the control sample was determined using the density method in accordance with ISO 1183–1:2025 (Method A). MFR strands were first weighed in air using an analytical balance (Mettler Toledo XSE105). Subsequently, the specimens were immersed in ethanol of known density at room temperature, and their apparent weight in the liquid was recorded with a density determination kit.

2.3.4. Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR)

Infrared spectroscopy analyses were conducted using a PerkinElmer Spotlight 400 FT-IR Spectrometer equipped with a Golden Gate diamond single attenuated total reflectance (ATR) cell. The MDOPE/LDPE films were delaminated and ATR-FTIR measurements were performed on the surface of the PUR layer. The spectra were recorded at a resolution of 2 cm^{-1} . Subsequently, the spectra underwent baseline correction and were normalized according to the maximum absorbance observed in the range 1760–1700 cm^{-1} .

2.3.5. Thermogravimetric analysis (TGA)

A thermogravimetric analyzer (TA Instruments Q500) was used to investigate the mass loss profiles of the compounded samples (7–8 mg). The samples were analysed in titanium crucibles using three successive dynamic steps: 10 $^{\circ}\text{C}/\text{min}$ from 25 to 180 $^{\circ}\text{C}$; 2 $^{\circ}\text{C}/\text{min}$ from 180 to 420 $^{\circ}\text{C}$; and 10 $^{\circ}\text{C}/\text{min}$ from 420 to 550 $^{\circ}\text{C}$ under a nitrogen atmosphere (60 mL/min). The heating rate was set at 2 $^{\circ}\text{C}/\text{min}$ in the temperature range of PURs mass loss (180–420 $^{\circ}\text{C}$) to increase the resolution of the analysis, accessing the PUR segments degradation steps, given the low concentration of PUR in the films (2 g/m²).

2.3.6. Rheological measurements

For the rheological analysis, the compounded samples were compression-molded into rectangular sheets with thicknesses of (1.0 $\pm$ 0.1) mm. Two layers were placed on top of each other to ensure sufficient material without voids. Any excess material was trimmed prior to measurement. Zero-shear viscosity values were calculated using the modified rheological model from Carreau (Equation (5)):

$$\eta = \frac{\eta_0}{[1 + (\lambda\dot{\gamma})^m]^{\frac{n-1}{m}}} \quad (5)$$

Where (η_0) represents the zero-shear viscosity, $\dot{\gamma}$ stands for shear rate, and λ , m , and n are the constants of this model [47].

An Anton Paar MCR 702e MultiDrive rheometer with a parallel plate geometry (diameter of $D = 25$ mm, gap between plates = 1 mm) was employed for oscillatory and rotational experiments. Oscillatory experiments were conducted under nitrogen atmosphere over an angular frequency (ω) range of 0.126–500 rad/s, at a temperature of 220 $^{\circ}\text{C}$, with an applied strain of 1 %. The viscosity curves of the samples were obtained at 220 $^{\circ}\text{C}$ within a shear rate range of 0.001–10 s^{-1} under a nitrogen atmosphere.

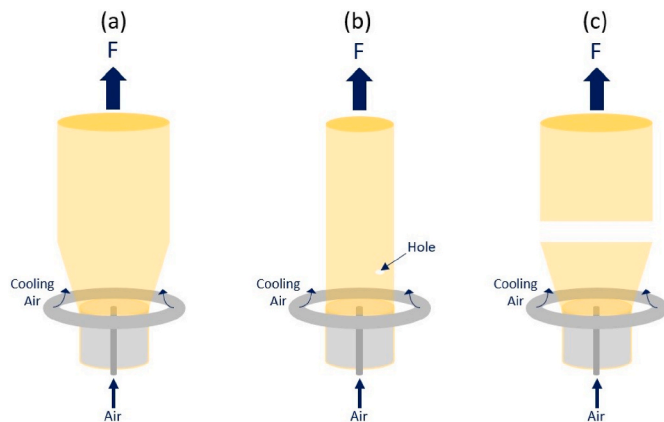


Fig. 1. Schematic representations of (a) a stable film-blowing process, and the two different instabilities observed for the 25 μm film setup: (b) width reduction and (c) bubble tearing.

2.3.7. Mechanical properties

Tensile tests of the recycled films were performed according to the ISO 527 standard, using a Zwick Roell Z020 testing machine equipped with a 100 N static load cell and a video extensometer. Test specimens (ISO 527 type 2) were cut from the films in both machine direction (MD) and transverse direction (TD), using a manual press (Ray-Ran) equipped with a cutting die. During the tensile tests, an initial pre-load of 0.1 N was applied, following a crosshead speed of 1 mm/min up to a strain of 1 % to determine the elastic modulus. Afterward, the test continued at 50 mm/min until the failure occurred. The tear force of the recycled films was measured according to the ISO 6383-1 standard using an Instron 5566 tensile test machine at a crosshead speed of 200 mm/min. The impact resistance of the films was measured according to the free-falling dart method (ASTM D 1709, method A) in an RDM FD-1709 falling dart impact tester.

2.3.8. Optical properties

The total light transmission, yellowing index (YI), and haze of the recycled films were measured according to the ASTM E1164 standard, using a HunterLab UltraScan VIS spectrophotometer in the transmittance mode.

2.3.9. Optical microscopy

A polarized optical microscope (Olympus DP27) equipped with a Linkam hot stage device was employed to assess the morphology of the PUR particles dispersed in PE. A temperature of 150 °C was set to melt the PE, facilitating the characterization of the crosslinked PUR particles. PUR particles were evaluated from five pictures of each sample, each representing a 1 cm² section of the films. The software Image-J was employed to measure the dimensions of the PUR particles. The reported number of gels corresponds to an equal surface area of 1 cm² for all samples.

3. Results and discussion

3.1. Characterization of the PE layers and control sample

Fig. 2a shows the viscosity of the laminate layers (MDOPE as the top layer and LDPE as the sealant) as a function of shear rate. MDOPE exhibits higher viscosity than LDPE at all shear rates, indicating that a higher molecular weight (Mw) grade was used in the production of the MDOPE layer [48]. Linear polyethylene grades with higher Mw are typically employed for MDOPE production to achieve films with enhanced physical performance [49].

The control sample (PU0), which is a blend of 50.3 wt% MDOPE and 49.7 wt% LDPE, presents higher viscosity than MDOPE. This higher viscosity can be attributed to crosslinking of PE chains in PU0, resulting from the additional high-temperature processing steps (compounding and film-blowing) occurred during the recycling process [50].

Zero-shear viscosity values were calculated using the modified rheological model from Carreau (Equation (5)) by fitting the curves as shown in Fig. 2a.

The zero-shear viscosities (η_0) of the MDOPE and LDPE layers, and control sample PU0 were 17823, 3988, and 20872 Pa s, respectively. The MFR values for the samples MDOPE LDPE, and PU0 were 0.52, 1.91, and 0.40 g/10 min, respectively. MFR values are consistent with polyethylene grades suitable for film production n [49].

The second heating cycle of the DSC thermogram for the laminate layers and PU0 sample is presented in Fig. 2b. Table A.3 in the Supporting Information (SI) contains the melting points, enthalpy of melting, and crystallinity percentages for all layers and PU0. Both MDOPE and LDPE layers display a single melting point, with MDOPE showing a higher melting temperature, indicative of a thicker lamellar structure resulting from its more linear chain configuration [51]. LDPE, with its lower melting temperature, is more appropriate for use as an inner sealant layer [52]. This observation is consistent with the density values, where MDOPE (931 kg/m³) exhibits a higher density than LDPE (921 kg/m³), further confirming its more linear chain structure [51]. The control sample, with a density of 927 kg/m³, lies between the two layers. The MDOPE layer also exhibits a higher melting enthalpy, reflecting a higher degree of crystallinity (Table A.3).

In contrast, PU0 exhibits two distinct melting points corresponding to the MDOPE and LDPE layers, suggesting that no substantial co-crystallization occurs in this blend due to the high degree of branching of LDPE [53]. However, the melting point of MDOPE in PU0 is slightly suppressed due to the presence of LDPE chains which may have either kinetic effect or thermal perturbations, which act as defects and lead to thinner lamellae [53].

Selecting the appropriate grades for MDOPE and LDPE layers can be guided effectively by evaluating the viscosity, density, and MFR values of each layer [54].

3.2. PUR characterization

3.2.1. ATR-FTIR

The FTIR spectra (400–4000 cm⁻¹) of the samples (PU0 – PU3 in Figure A.9) indicates that PU1 is a polyether-based PUR, with a strong band at 1098 cm⁻¹, referring to the C-O stretching mode vibration in ethers. PU2 and PU3 most likely consist of polyester-based PUs since they do not present absorption peaks associated with ether bands [55]. Fig. 3a presents the ATR-FTIR spectra of the PU0-PU3 sample in the region related to the carbonyl region (1620–1780 cm⁻¹), emphasizing their different compositions and hydrogen-bonding (H-bond) characteristics.

FTIR is a very sensitive technique for detecting hydrogen bonding development in the carbonyl region of PURs. The maximum absorption for the carbonyl group begins around 1730–1750 cm⁻¹ for free, non-H bonded carbonyls in the PURs soft segments, and around 1720–1740

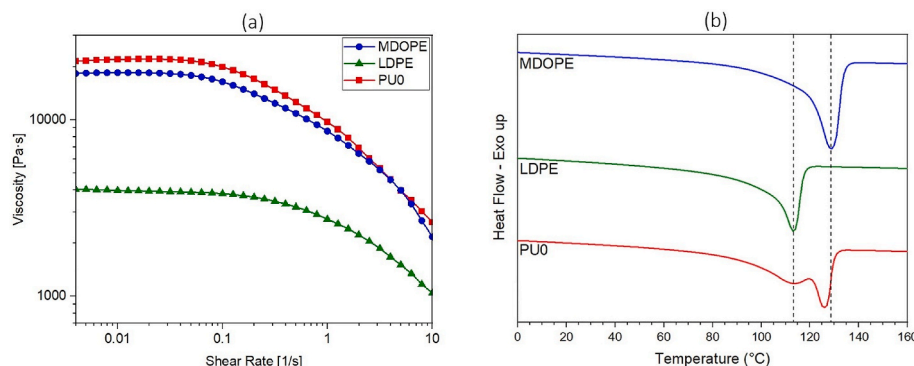


Fig. 2. (a) Viscosity curves of PE layers and the control sample as a function of shear rate. (b) Second heating cycle DSC thermograms of the PE layers and the control sample.

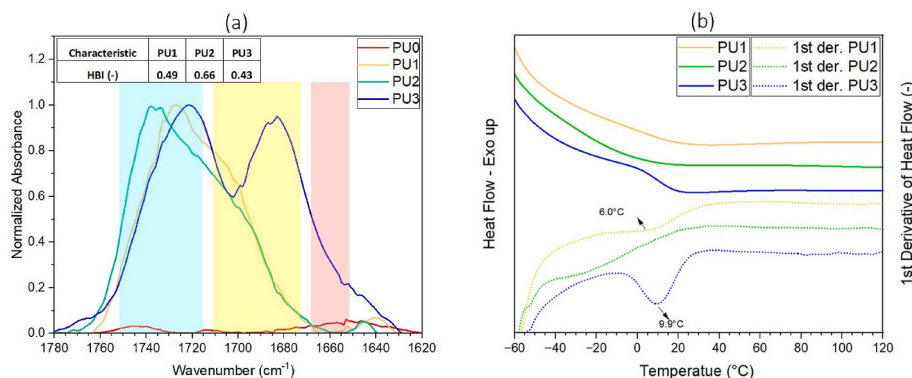


Fig. 3. (a) Normalized absorbance in the carbonyl region of the FTIR spectra for PU0-PU3 samples. The hydrogen bond indexes (HBI) is depicted in the inset table. The red region represents the ordered H-bonded carbonyl groups, the yellow indicates disordered H-bonded groups, and the blue corresponds to free urethane carbonyl groups. (b) DSC results for the second heating cycle of the extracted PUR samples.

cm^{-1} for free urethane carbonyl groups in the PURs hard segments. This band shifts to $1680\text{--}1720\text{ cm}^{-1}$ for H-bonded carbonyl and urethane carbonyl in both soft and hard segments in a disordered phase and ultimately shifts to $1640\text{--}1680\text{ cm}^{-1}$ for H-bonded carbonyl of urethane groups in an ordered phase of hard segments [33,56–59]. Additionally, carbonyl bands associated with urea groups forming H-bonds appear in the range of $1620\text{--}1650\text{ cm}^{-1}$ [33]. Therefore, the carbonyl absorption region was divided into three segments: (i) ordered H-bonded carbonyl groups (around 1662 cm^{-1} only in case of PU3); (ii) disordered H-bonded carbonyl groups (around 1706 cm^{-1} for PU1, 1712 cm^{-1} for PU2, and 1684 cm^{-1} for PU3); and (iii) free, non-H-bonded carbonyl groups ($1723\text{--}1739\text{ cm}^{-1}$) [33,58]. The FTIR spectra of the carbonyl region were deconvoluted based on these three distinct segments, and the area of each assigned peak was calculated. The deconvolution results for each absorption band are presented in Figure A.10 and Table A.4. Thereafter, the hydrogen bond index (HBI) was determined using Equation (6) [29]:

$$HBI = \frac{S_{\text{bonded}}}{S_{\text{bonded}} - S_{\text{free}}} \quad (6)$$

Where S_{bonded} represents the area of the H-bonded carbonyl peak ($1660\text{--}1712\text{ cm}^{-1}$), and S_{free} represents the area of the free carbonyl peak ($1720\text{--}1740\text{ cm}^{-1}$). HBI values for all samples are presented in Fig. 3a.

The development of hydrogen bonding networks within the hard segments or between hard and soft segments, and the microphase separation between hard and soft phases determine the morphology and properties of PURs [33,60]. Analyzing the FTIR spectra of the samples, it is shown that PU3 exhibits a distinct absorption pattern compared to PU1 and PU2. PU3 shows three strong absorption peaks corresponding

to two H-bonded peaks at 1662 and 1684 cm^{-1} (present in the ordered phase in hard segments and disordered phase within hard-soft segment, respectively) and free, non-H bonded urethane carbonyl groups. In contrast, PU1 and PU2 show a single peak in the range of free urethane carbonyl, while only a shoulder towards the disordered H-bonded carbonyl groups region between soft and hard segments is observed. The ordered H-bonded carbonyl groups in the hard phase of PU3, as indicated by the peak 1662 cm^{-1} , suggest that some of the hard segments are incorporated into the hard phase and developed H-bond with other hard segments [31,61], leading to phase separation between the hard and soft microphases [57]. A small shoulder in the range of $1706\text{--}1712\text{ cm}^{-1}$ for PU1 and PU2 indicates the presence of weakly H-bonded hard segments [61]. Considering these peaks for PU1 and PU2, it is shown that a fraction of hard segments is dispersed within the soft segments, resulting in a lower level of phase separation for these samples [57]. The proposed morphology model of PU1 and PU2 is shown in Fig. 4a, and that of PU3 is depicted in 4b. For simplicity, the isocyanate functionality, polyol molecular weight, and isocyanate index are assumed to be equal for all three PURs. The HBI values indicate that PU2 has a higher ratio of H-bonded urethane carbonyls compared to PU1. However, PU3 presents the lowest ratio of H-bonded urethane carbonyls, while the hard segments in PU3 show ordered hydrogen bonds compared to PU1 and PU2, indicating a morphology consisting of packed and phase-separated hard segments for PU3, as evidenced by the strong absorption related to H-bonded urethane carbonyls at lower wavenumbers. The type and symmetry of the isocyanate, the type and molecular weight of the polyol, and the ratio between these PURs constituents (isocyanate index) are important factors affecting the H-bond network in the final PUR structure. The linear and planar isocyanates enable more H-bonds and higher ordering of the hard segments in the hard domains [57,61].

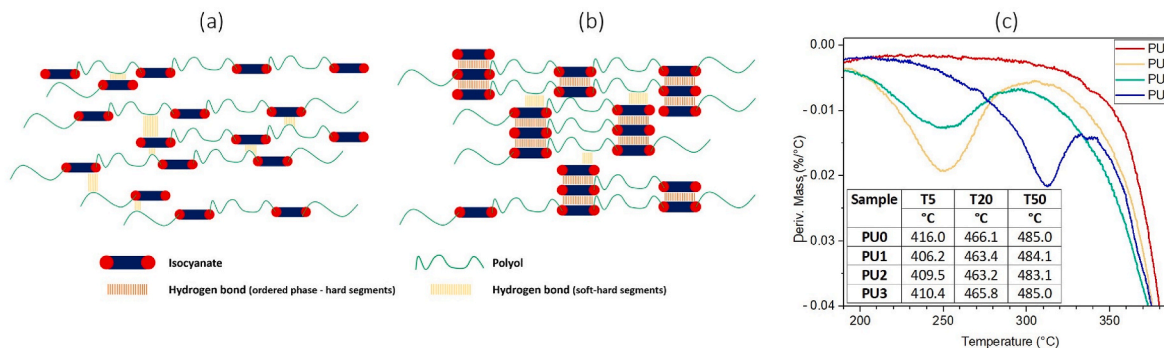


Fig. 4. (a) Proposed morphological models of PU1-PU2. (b) Proposed morphological models of PU3. (c) First derivative of TGA thermograms during the initial stage of mass loss (from 190 to 390 °C). In the inset table: the different amounts of mass loss (5, 20, and 50 wt%– T5, T20, and T50, respectively) during the thermal decomposition of the samples.

Additionally, increasing the isocyanate index results in higher amount of H-bonds in the final PUR [62,63]. A higher molecular weight polyol or polyols presenting lower compatibility with the hard segments lead to a higher degree of phase separation and higher HBI values [64].

Concluding the FTIR investigation, the PU1 and PU2 samples show small peaks in the range characteristic of urea group absorption ($1630\text{--}1650\text{ cm}^{-1}$) [33]. In the case of PU3, this peak likely overlaps with that of the ordered H-bonded urethane group and is therefore not distinguishable. The formation of urea groups is a known side reaction during PUR polymerization, resulting from the reaction of isocyanates with water [65,66].

3.2.2. DSC

The DSC results obtained during the second heating cycle for the extracted PURs is shown in Fig. 3b. Both the heat flow and the first derivative of heat flow for the PUR samples are shown. PURs presenting high miscibility between the hard and soft segments typically exhibit a single glass transition temperature (T_g), positioned between the T_g values of the hard and soft segments [67,68]. Among the samples studied, only PU3 displayed a well-defined glass transition at $T_g = 9.9\text{ }^\circ\text{C}$, as highlighted by the peak observed in the first derivative of heat flow of this sample. In contrast, PU1 and PU2 exhibited T_g in a broad range of temperature, with estimated average T_g values lower than that of PU3. The broad transition temperatures observed for PU1 and PU2 is attributed to the extensive degree of mixing between the hard and soft segments, being consistent with the FTIR results shown in Fig. 3a [69,70].

3.2.3. TGA

TGA provides insights regarding PUR composition and microstructure, as the thermal degradation of PUR depends on the hard and soft segment types, isocyanate index, and degree of phase separation [30]. Figure A.11 presents the TGA thermograms (mass versus temperature) of PU0-PU3 of samples from 25 to $550\text{ }^\circ\text{C}$, while Fig. 4c depicts the first derivative (DTG) of the mass change as a function of the temperature curves during the initial stage of mass loss (from 190 to $390\text{ }^\circ\text{C}$) for PU0 - PU3 samples.

Under a nitrogen atmosphere, PU0 (PE) shows a single step of mass loss starting at $350\text{ }^\circ\text{C}$ [71], as observed in Fig. 4c. For the samples containing PUR adhesives (PU1-PU3), the mass loss occurs in two or three steps, starting around $200\text{ }^\circ\text{C}$ [30]. The thermal degradation of PUR initiates with the breakdown of urethane bonds from the hard segment, and the onset temperature and magnitude of this stage depend on the type of isocyanate and polyol, as well as their ratio. A higher content of the soft segment decreases the rate and intensity of the mass loss observed during this first degradation step [34]. In addition, urethane groups presenting low amount of hydrogen bonds are less stable and degrade faster [32]. The second and third DTG peaks are related to the decomposition of soft segments which occur at a slower rate than the first step [30]. PU3, presenting a higher content of ordered hard segments shows mass loss at higher temperatures [30,72]. PU1 and PU2 show a similar first DTG peak, starting at around $195\text{ }^\circ\text{C}$ with a maximum mass loss rate at $250\text{ }^\circ\text{C}$. However, PU1 shows a higher rate of mass loss, indicating a possible higher content of hard segments. Conversely, PU3 demonstrates a diverse thermal decomposition behavior, starting at a higher temperature compared to PU1 and PU2, and showing the first DTG peak as a shoulder beginning at $210\text{ }^\circ\text{C}$ and the second DTG peak with a maximum at around $311\text{ }^\circ\text{C}$. These results suggest that PU3 presents more stable hard domains, such as urethane formed from alkyl-isocyanate and alkyl-polyols instead of aryl-isocyanate and aryl-polyol [32]. Furthermore, increasing the crosslinking density or the isocyanate index improves the PURs thermal stability since more energy is required to breakdown the PUR network [30].

The temperatures at which the samples presented 5, 20, and 50 wt% of mass loss during the TGA test (T_5 , T_{20} , and T_{50} , respectively) were

determined and are shown in the inset of Fig. 4c. PU1 starts to degrade earlier in the first step than PU2, for T_5 and T_{20} . However, at a higher temperature, the DTG degradation peak of PU1 happens at a higher temperature than the one for PU2. This behavior is possibly related to the higher content of isocyanates and different types of polyols in the PUR's structures [34,72]. PU3 presents a higher thermal stability compared to PU1 and PU2, conceivably due to its ordered hard segment morphology, the content of hydrogen bonds in the hard phase, and a higher crosslinking degree.

3.3. Processability during closed-loop mechanical recycling

During the film-blowing process, numerous bubble instabilities can occur, depending on the polymer molecular structure and processing parameters, such as helical instability, draw resonance, bubble sagging, FLH instability, bubble tearing and rupture, and width reduction, limiting the production [42,73,74]. During the shredding and compounding steps of MM-PE laminates, no abnormal conditions or differences were observed. However, variations in the film-blowing processability of the samples were noticed. Detailed processability criteria are presented in Fig. 5.

From Fig. 5, it is evident that PU3 imposes a higher load on the extruder drive compared to the other samples. PU1 leads to an extruder load similar to the one of PU0, while PU2 exhibits the minimum extruder load. PU1 and PU3 show similar melt pressures, and higher than PU0. Both extruder load and melt pressure are influenced by the melt viscosity and pressure of the polymer at the extrusion condition [75]. Another important parameter for the film production industry is the melt pressure build-up. It is common to incorporate a melt filtration step in the film-blowing line to remove contaminations and gels. However, a faster melt pressure build-up leads to more frequent filter replacements to prevent damage to the line's head and die. Therefore, lower melt pressure build-up is preferred in the film production industry. Among the samples, PU3 shows the highest melt-pressure build-up, followed by PU1, PU2, and PU0.

During the film-blowing process, maintaining the bubble stability is essential for a stable process. In this study, only bubble tearing and width reduction were observed as instabilities, as shown in Fig. 1. From Fig. 5, it is observed that PU3 poses substantially more challenges during the filmblowing process, presenting 39 bubble tearing events. PU2

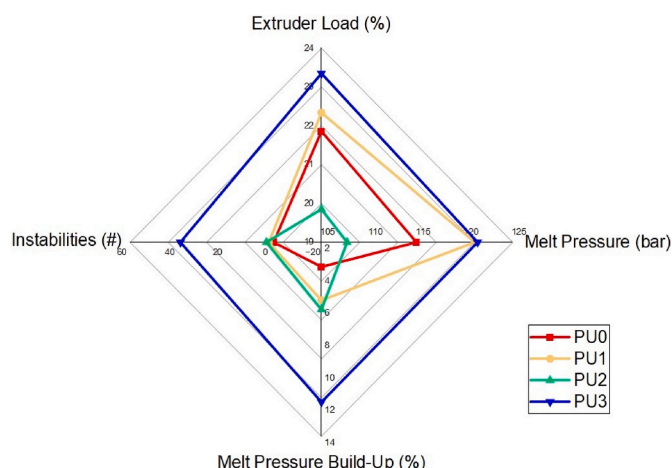


Fig. 5. Processability data for the samples during the production of $25\text{ }\mu\text{m}$ films. The melt pressure build-up is calculated using Equation (6), representing the increase in melt pressure relative to the average melt pressure shown in this figure. Instabilities are quantified as the summation number of B-R over a 1-h processing time. The extruder load refers to the percentage of the torque used during processing relative to the machine's maximum torque, which is 150 Nm. In the film-blowing process, lower levels of instability, extruder load, melt pressure, and melt pressure build-up are preferred.

showed 3 instabilities, PU1 had 2, while no instabilities were recorded for PU0. Based on FTIR and TGA analyses, PU3 exhibited the characteristic features of a PUR with a more ordered hydrogen-bonding network within the hard segment, and a higher crosslinking density compared to PU1 and PU2. These structural properties suggest that PU3 particles are more stiff and rigid, which may promote bubble rupture and hole propagation. The visual quality of these films is presented in Figure A.12.

3.4. Rheological assessment

3.4.1. Linear viscoelastic measurement

The linear viscoelastic properties in terms of storage and loss moduli as a function of angular frequency of PU0-PU3 samples at 220 °C are depicted in Figure A.13. PU3 exhibits storage and loss moduli comparable to those of PU0. However, PU1 and PU2 show a slight decrease in their storage and loss moduli compared to PU0. Given the similar PE composition across all the samples, this change can be attributed to the structural difference, including types of polyol and isocyanate, hydrogen bond interactions, and isocyanate index among the PUR adhesives and their dispersed particle morphology which is discussed in section 3.6.

The dynamic intersection of storage and loss moduli curves, and the dynamic intersection frequency values of the samples can be seen in Fig. 6a. A higher value of the dynamic intersection or an increase in the crossover frequency indicates a less homogenous system [76,77]. As such, PU3 shows the lowest degree of homogeneity among the samples, while PU1 demonstrates the highest level of homogeneity among samples containing PUR adhesive and its homogeneity level is similar to PU0 (Control sample). PU2 is positioned between PU1 and PU3 based on intersection frequency. The homogeneity of these systems is related to the chemical structure of the PUR adhesives. For instance, polyether polyol-based PUR is expected to have a better affinity with PE. This may result in a more regular morphology for the PUR dispersed phase than polyester polyol-based PUR [78]. From this result it is expected that PU1 presents molecular segments with lower surface energy, thus showing better affinity with PE chains. In contrast, PU3 contains more polar groups, and it is less compatible with the non-polar PE chains. As previously described, PU1 is based on a polyether polyol and PU3 is based on a polyester polyol. Also, PU3 showed a higher proportion of hydrogen bonds within the hard phase, which makes PU3 more heterogeneous in the recycled laminate leading to a higher number of instabilities during the film-blowing process. The shift of crossover frequencies to higher values corroborates with the poor processability of PU3 during the film-blowing, as it leads to more bubble rupturing [79].

The shear rate in the extruder can be calculated using the general formula for the single screw extruders (Equation 7):

$$\dot{\gamma} = \frac{N_s D_s \pi}{H_s} \quad (6)$$

Where N_s represents the number of screw rotations per second, D_s , the screw diameter, and H_s , the screw flight depth [80]. During the extrusion of all samples, an average shear rate of 23.5 s^{-1} was calculated using Equation 7. The $\cot \delta$ parameter, derived from the oscillatory rheological analysis, is equal to the ratio of storage-to-loss moduli (inverse of $\tan \delta$ parameter) and indicates the melt rigidity properties of polymers [81]. The $\cot \delta$ of the samples at the frequency of 23.5 s^{-1} were calculated considering the validity of the Cox-Merz rule since the content of PUR is low and are shown in Fig. 6a [82]. The value of $\cot \delta$ of each sample can be found in Figure A.13. A higher value of $\cot \delta$ is favorable for producing a stable bubble during the film-blowing process [79]. PU0 shows the highest value of $\cot \delta$, indicating high melt rigidity and bubble stability. PU1 shows a $\cot \delta$ value similar to that of PU0, and PU2 and PU3 show lower $\cot \delta$ values, consistent with the processability results.

3.4.2. Zero-shear viscosity

Viscosity values, particularly at low shear rates, are associated with the molecular characteristics of the polymer chains. The value of viscosity at the Newtonian plateau, referred to as zero-shear viscosity (η_0), is indicative of the degree of entanglement, and consequently, the melt strength of the material [83,84]. Fig. 6b presents the viscosity of PU0-PU3 samples as a function of the shear rate. From Fig. 6b, one can observe that PU0 and PU1 showed similar viscosity as a function of the shear rate.

In contrast, PU2 and PU3 show lower viscosities across the entire shear rate interval, especially in the Newtonian plateau, indicating lower zero-shear viscosities. For a more accurate comparison, the zero-shear viscosity values were calculated using the modified Carreau rheological model (Equation (5)) by fitting the curves in Fig. 6b. From this fitting PU1 exhibits a higher value of zero-shear viscosity than PU0. For PU2 and PU3, the zero-shear viscosities drop by 28 % and 13 %, respectively, in comparison to PU0. The zero-shear viscosity value for each sample can be found in the table of Fig. 6b. Higher zero-shear viscosity indicates higher melt strength and higher stability during film-blowing [85,86]. PU2 and PU3 appear to act as a less viscous component for PE chains, facilitating the macromolecules' mobility [87,88]. This behavior is similar to the viscosity profile observed in incompatible polymer blends caused by weak interfacial interactions [89]. At high shear rates, the shear-thinning behavior of PE is dominant, and all the samples represent similar viscosity values.

At the extrusion shear rate, the melt viscosity of the samples were calculated by fitting the modified Carreau rheological model (Equation (5)) and are shown in the table of Fig. 6b. PU1 exhibits the highest melt

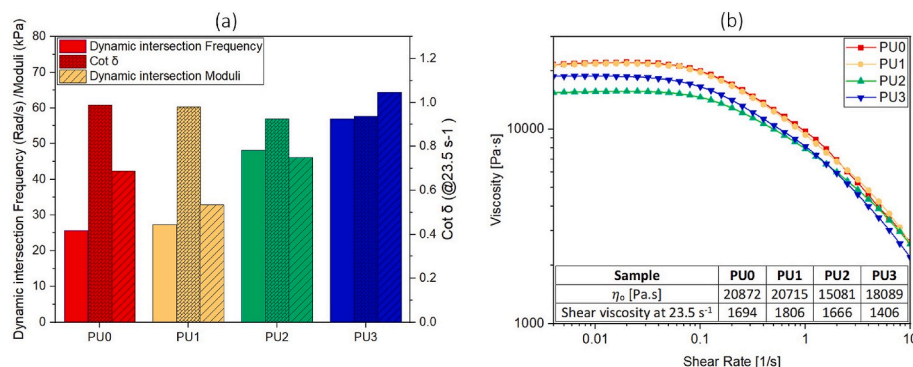


Fig. 6. (a) Linear viscoelastic properties (Including dynamic intersection frequency and moduli, $\cot \delta$ of PU0 and samples containing different PUR adhesives (PU1-PU3)) at 220 °C, (b) Viscosity curves as a function of the shear rate, the zero-shear viscosity values, and the shear viscosities at 23.5 s^{-1} of the studied samples at 220 °C.

viscosity under the extrusion conditions, followed by PU0, PU2, and PU3. Notably, viscosity itself does not elucidate the variation in the extruder load in the processability of the samples during the production of 25- μm films (Fig. 5). However, samples with higher melt pressure build-up and melt pressure, such as PU3, pose a higher load on the extruder motor, which can be attributed to the deposition of crosslinked PUR particles as contaminants on the screen filter.

3.5. Mechanical properties

Fig. 7 depicts the tensile modulus, tensile strength, elongation at break, dart drop resistance, and tear force results for the studied samples, both in MD and in TD. To develop new applications for mechanically recycled plastics in the film industry, high values for dart drop impact resistance, elongation at break, tensile strength, and tear strength are preferred. As for tensile modulus, either low or high values may be desirable depending on the specific end-use application.

The average tensile modulus of PU2 in the MD is slightly higher than that of PU0, while it is lower for PU1 and PU3. This may be attributed to the presence of stiffer PUR particles in PU2, which have a stronger interface with the PE matrix and act as reinforcing agents within it [90, 91]. In the TD, the tensile modulus follows a similar trend as the MD. However, the absolute TD values are higher, due to the strong orientation of PE chains in the MD at high DDR values [92]. At high DDR, the PE melt undergoes strong stretching in the MD and is subsequently quenched immediately after exiting the die. This rapid solidification preserves the chain orientation in the MD, while the crystalline lamellae tend to stack preferentially perpendicular to this direction [92–94]. Such a microstructural arrangement reduces stiffness along the MD but enhances resistance to deformation in the TD, resulting in a higher modulus in the latter [92–94]. In the TD, the films behave as an iso-strain system, where the crystalline regions, showing higher modulus than the amorphous phase, dominate the overall tensile response [92]. Therefore, the increase in modulus for PU2 is less pronounced in the TD compared to the MD, as the TD already exhibits a higher initial modulus. In the case of PU3, the decrease in modulus is more significant in the TD than in the MD, likely due to poor stress transfer between the PE and PUR phases [95]. Since the crystalline regions with higher modulus dominate the tensile behavior in the TD, the material experiences higher stress and its microstructure becomes more sensitive to defects. Finally, PU1 shows a reduction in modulus in the TD similar to that observed in the MD.

For all samples containing PUR adhesive, the tensile strength in both MD and TD shows a slight but statistically significant decrease, associated with poor stress transfer between the PE and PUR phases [90]. Only PU2 in the TD shows tensile strength comparable to that of PU0. Overall, considering tensile strength in both MD and TD, all PUR adhesives behave as incompatible contaminants, with limited stress transfer between the PE matrix and PUR domains. This results in films with lower

mechanical properties compared to PU0 [90,96].

Regarding the elongation at break, PU1, PU2, and PU0 show similar values in the MD. In the TD, PU2 and PU0 have similar elongation at break values, and PU1 shows a slight reduction in this property. In both MD and TD, PU3 demonstrates a significant reduction in elongation at break (by 55 % and 53 % in MD and TD, respectively). The presence of PU3 particles in the PE matrix decreased the ductility of the PE film by introducing stress concentration points and a weak interface between the components of the blend [90,96,97].

Similar to the tensile test results, the tear force in the TD is higher than in the MD, due to higher chain orientation in the MD [98,99]. PU2 shows a slightly higher tear force compared to PU1 and PU0 in MD. In addition, PU3 recorded the lowest tear force among the samples, 16 % lower than PU0. In the TD, PU2 and PU0 demonstrate similar values of tear force, with PU1 slightly lower than those two samples. Finally, PU3 resulted in a 27 % reduction in the TD tear force compared to PU0, mainly related to the stress concentration caused by the existence of non-compatible PUR particles in the film structure, which promotes the initiation and propagation of cracks [98–100].

Dart drop impact results are substantially more sensitive to the presence of PUR particles compared to other mechanical properties. The dart drop impact resistance of the samples decreased by 9, 40, and 72 % for PU1, PU2, and PU3 respectively, compared to PU0. The presence of PUR particles resulted in more brittle films with lower deformability, likely due to the stress concentration around the PUR particles and less stress transfer in the PUR/PE interface [96].

3.6. Colorimetric and PUR adhesive particle size

Fig. 8a represents the optical properties of all the samples in terms of total light transmission, YI, haze, and gel size.

All samples exhibited total light transmission values above 89 %, indicating high transparency. However, PU1, PU2, and PU3 showed a statistically significant decrease in light transmission, due to the presence of PUR particles. PU1 presented the highest value of light transmission, followed by PU2, and finally PU3. Typically, higher phase separation between the hard and soft segments of PUR results in lower light transmission, corroborating the results observed for PU3 [101].

Considering the YI, PU1 shows the highest YI value. The yellowing effect is a result of PUR degradation during the thermal process [101, 102]. Isocyanates containing aromatic moieties lead to more yellowing compared to aliphatic ones [103]. The TGA results of PU1, previously discussed (Fig. 4c), indicated that this sample degrades at lower temperatures and at a faster rate compared to PU2 and PU3 at the extrusion temperature, which leads to the higher YI value.

Irregularities originating from the polymeric film production process, rheological properties, and contamination cause the scattering of the light, observed as film haziness [104,105]. The presence of cross-linked PUR particles increased the haziness of PE films (Fig. 8a). The

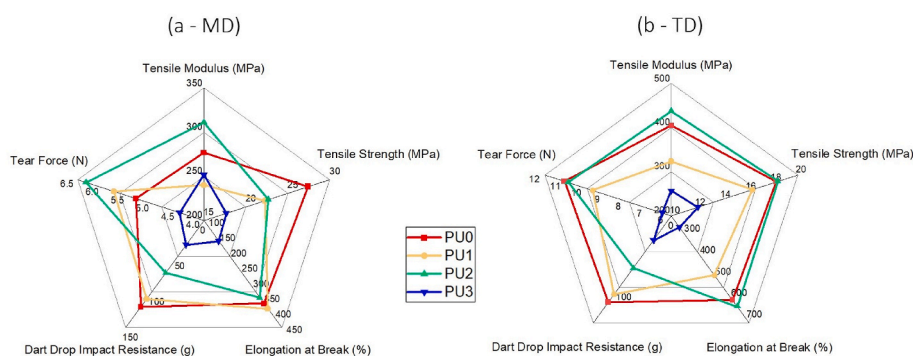


Fig. 7. Mechanical properties of PU0-PU3 for the 55 μm film setup: (a) Mechanical properties of MD and dart drop impact resistance of samples and, (b) Mechanical properties of TD. Each graph uses a different scale. Since dart drop impact resistance is not direction-dependent, it is represented in both the MD and TD graphs.

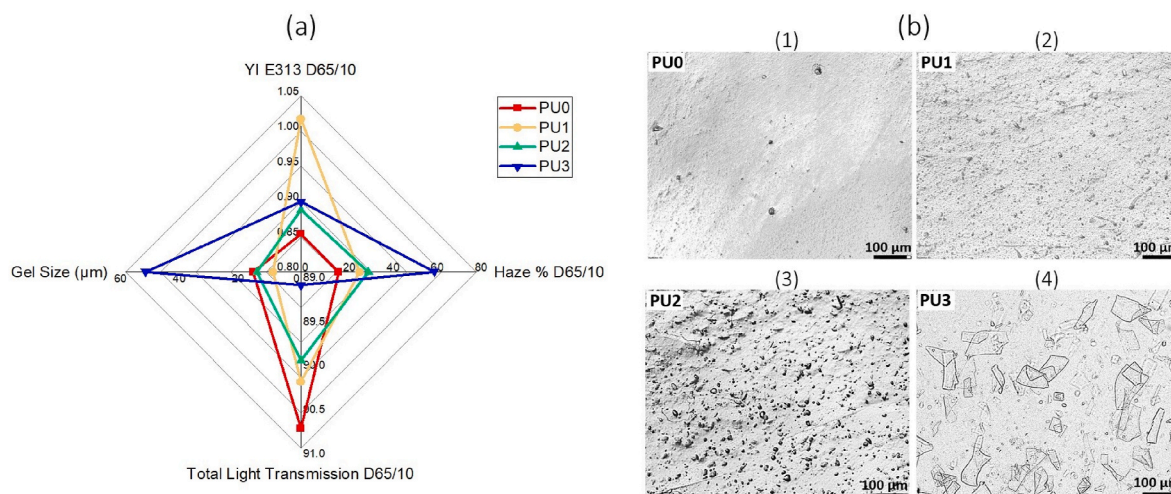


Fig. 8. (a) Colorimetric properties of PU0-PU3 including total light transmission, YI, haze, and average gel size for 55 μm film setup. Higher total light transmission leads to greater transparency in packaging films. Lower levels of haze and YI are preferred in the packaging industry. (b) Polarized microscopy pictures of the blown films with thickness of 55 μm : 1. PU0; 2. PU1; 3. PU2; and 4. PU3; The scale in the pictures is 100 μm .

haze value of PU3 is considerably higher than PU2, PU1, and PU0, related to the presence of larger particles in PU3 (average particle size of 53.11 μm). The haze results correlate with the average PUR particle sizes, as shown in the polarized microscopy images of the samples (Fig. 8b). Sample PU0 consists of pure PE and also present gel particles, resulting from the degradation through crosslinking of PE chains during the extrusion process [106]. The average gel size of PU0 is larger than that of PU1 (16.38 and 9.53 μm , respectively), whereas PU1 contains a high number of round and small PUR particles, leading to a lower average gel size. The numbers of gel particles found in each sample are: 32 for PU0, 415 for PU1, 878 for PU2, and 97 for PU3.

PU2 also displays rounded PUR particles, however larger, while PU3 presents large and sharp PUR particles. In sample PU3, a higher amount of ordered hydrogen-bonds within the hard segments results in bigger PUR particles in the PE matrix, causing more instabilities during the film-blowing process compared to PU1 and PU2. PU1 particles are smaller than PU2, possibly due to the composition of PU1 which contains polyether polyol which has better affinity with PE chains. According to the group contribution theory [107], the solubility parameter of propylene glycol (a representative polyether) is.

17.65 $(\text{J}/\text{cm}^3)^{0.5}$, while is 19.94 $(\text{J}/\text{cm}^3)^{0.5}$ for a polyester based on adipic acid and diethylene glycol. Since PE has a solubility parameter of 17.1 $(\text{J}/\text{cm}^3)^{0.5}$, which is closer to that of polyether-based polyols, polyether-based polyols exhibit better affinity with the PE matrix [78, 107].

3.7. Compilation of the results

Table 1 compiles all the data related to the processing and characterization of PE-containing PUR adhesives in the closed-loop film-blowing process.

PU3 results in higher heterogeneity, and it reduces the zero-shear viscosity, or melt strength, which are the main reasons for the poor processability of this sample. Based on the FTIR data, PU3 presents, large quantity of hydrogen bonding in the hard phase. PU1, showing a lower amount of hydrogen bonds, has a lower degree of phase separation and a better affinity is expected between the dispersed PUR and the PE matrix, with a higher zero-shear viscosity, resulting in a more stable film-blowing process, comparable to PU0. It is worth mentioning that PU2 demonstrates satisfactory stability during the film-blowing process. However, mainly due to its low zero-shear viscosity, it shows more instabilities than PU0 and PU1. PU1 and PU2 present good mechanical and optical properties, similar to those of PU0. A small amount of PU3

Table 1

Comparison of PUR-based samples in terms of structure, processing, and performance.

	PU0	PU1	PU2	PU3
PUR H-bond development	0	H-bonded carbonyl groups (soft-hard segments)	H-bonded carbonyl groups (soft-hard segments)	Ordered phase (hard segments)
PUR's amount of ordered hard segments and crosslinking ^a	0	Low	Low	High
Process stability	+++	+++	++	+
Mechanical performance (Toughness)	+++	+++	++	+
Optical properties (Light transmission and haziness)	+++	+++	++	+

^a Based on FTIR and TGA analyses.

adhesive in the PE matrix (app. 2 wt%) results in poor mechanical and optical performance. However, the same amount of PU1 yields mechanical and optical properties similar to PU0.

4. Conclusion

In this study, the effect of PUR adhesives on the recyclability of MM-PE laminated films was assessed during a closed-loop film-to-film mechanical recycling process. It is shown that the type of PUR adhesive will affect the recyclability of MM-PE films, leading to higher melt pressure build-up and more instabilities during the film-blowing process. Rheological investigation of recycled MM-PE laminates containing PUR adhesives proved to be an effective method to evaluate their processability and large number of instabilities during the film-blowing process.

PURs introduced heterogeneities in the PE matrix, shifting the dynamic intersection of the storage and loss moduli, as well as the frequency at this intersection, to higher values. Furthermore, the presence of the PURs may reduce the zero-shear viscosity, lowering the melt strength and causing instabilities, affecting the film-blowing process. PUR adhesives showing higher degree of separation between hard and soft phases with large amount of H-bonding between hard segments, inflict a less stable film-blowing process. PURs strongly affect the

toughness of the PE film by decreasing the dart drop impact resistance and elongation at break. The thermal degradation of PURs caused yellowing in the recycled films, and larger PUR particles led to a higher haziness.

The recyclability of MM-PE laminated films containing PUR adhesives can be systematically assessed by investigating the physicochemical characteristics of the PUR adhesive by FTIR and TGA analysis. The methodology and results of this study can be employed as a framework for assessing the recyclability of MM-PE laminates containing PUR adhesives.

This study highlights how existing Design for Recycling (DfR) guidelines for PE-based flexibles could be further refined down to the level of the chemistry used in adhesives for MM-PE. Overall, maximizing recycling efficiency requires a holistic approach in which all components, especially those present in minor quantities, such as labels, adhesives, lacquers, inks, and pigments, are carefully selected and assessed for compatibility with the intended recycling process and each other. Only through such comprehensive material design can true circularity in flexible packaging be achieved.

CRediT authorship contribution statement

Ali Ebrahimi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Francis Pardal:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis, Data curation. **Marion Jarnoux:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Marc Defoin:** Visualization, Supervision, Project administration, Funding acquisition, Formal analysis. **Livia Mesquita Dias Loiola:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Kim Ragaert:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Funding acquisition, Formal analysis. **Rudinei Fiorio:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ali Ebrahimi reports financial support was provided by Bostik Smart Technology Center. Reports a relationship with that includes: Has patent pending to. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polymertesting.2025.109010>.

Data availability

Data will be made available on request.

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